INCH-POUND

MIL-L-80081D
9 AUGUST 1990
SUPERSEDING
MIL-S-80081C
18 May 1981

MILITARY SPECIFICATION

LATHES, WOODWORKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers standard bed and fixed gap bed woodworking lathes, with equipment and accessories specified herein. The requirements set forth in this specification and the completed ordering data (see paragraph 6.2), establishes the Government's minimum requirements for each procurement.

1.2 Classification. The lathes shall be of the following type, sizes and styles. The type, size and styles to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Direct (motor headstock) drive, 4 speeds
  Size 12 - 12 inches swing over bed

Type II - Belt drive
  Size 12 - 12 inches swing over bed
    Style 1 - infinitely variable speed
    Style 2 - 4 stepped speeds
  Size 14 - 14 inches swing over bed
    Style 1 - infinitely variable speed

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be ad dressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3220

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.) Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

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(OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI O1.1 - Safety Requirements for Woodworking Machinery.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete lathe shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The lathes shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The lathes shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.4.1), metric (see 6.4.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the sanders shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The lathes and their components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All electrical, mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

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3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the lathes, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI 01.1. If a conflict arises between 29 CFR 1910 and ANSI 01.1, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial lathes supplied to the commercial market shall be provided for the point of operation and all other parts of the lathes that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The lathes shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the lathes.

3.2.8 Environmental protection. During the construction of the machine, the manufacturer shall comply with all applicable Federal, state, and local statutes in effect at the point of manufacture on the date of the contract which regulate the use of materials hazardous to the ecological system. During its operation, the machine shall comply with all applicable Federal statutes in effect on the date of the contract which regulate the emission of hazardous materials to the environment.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective parts between end items of the same model. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.3 Construction. The lathes shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the lathes to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

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3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. Any material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the lathes and their related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All gears shall have a surface finish, and toughness that will enable the gear train to transmit full-rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. Each lathe shall consist essentially of a base, bed, headstock, tailstock, drive and such other components as are described herein.

3.4.1 Base. The lathe components shall be supported by a base with legs of pedestal or cabinet construction. The base shall have provisions for bolting to the floor or foundation.

3.4.2 Bed. The bed shall be of such mass and rigidity to support the headstock and such other components with workpieces mounted thereon. The bed shall have ways on the top to provide longitudinal adjustments of the toolrest holder and the tailstock. When specified (see 6.2), the type II, size 12 lathe bed shall be constructed with a gap to provide swing capacity in accordance with table I. The swing capacity over the bed and between centers shall be not less than the requirements of table I. When a hand feed carriage

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is specified in 3.4.5.1, the bed shall have an inverted V-way on the front of the bed, and have a rack gear which will mesh with the manually controlled pinion to provide longitudinal carriage movements.

3.4.3 Headstock. The headstock shall support the spindle, workpiece mounting and driving devices, and the workpieces mounted thereon. The headstock on type I lathes shall totally enclose the motor and the spindle speed selection mechanism. The spindle and spindle speed components attached to the spindle shall be supported by the headstock on ball or roller bearings of a type, class and size suitable for the application. The headstock of type II lathes shall be fitted with a removable cover or guard over the spindle driving device allowing access to the covered parts.

3.4.3.1 Headstock spindle. The headstock spindle shall be hollow to provide for removal of the spindle center. Spindles shall have means for mounting of face plates on both ends. The spindle end, facing the tailstock, shall have a Number 2 Morse taper. Spindle speeds shall conform to table I. The spindle shall either be threaded on each end or threaded on the end facing the tailstock with an adapter mounting on the opposing end, for receiving faceplates. The spindle shall be supported by spindle bearings of either the tapered roller or the ball type. Tapered roller spindle bearings shall have means of adjustment.

3.4.4 Tailstock. A tailstock shall be furnished on each lathe. The tailstock shall be mounted to support the free end of the workpiece. The position of the tailstock shall be adjustable to accommodate workpieces having any length within the center to center dimension specified in table I. On lathes having a hand feed carriage, the tailstock shall incorporate a set-over feature to facilitate taper turning of workpieces.

3.4.4.1 Tailstock spindle. The tailstock spindle shall be horizontally adjustable, by means of a handwheel-operated screw, for a distance of not less than 2-1/4 inches. The tailstock spindle shall receive the shank of centers or other adapters having a Number 2 Morse taper. The tailstock spindle shall have a locking device to retain the spindle in horizontal adjustment while supporting the free end of the workpiece.

3.4.5 Toolrest holder. Lathes shall be furnished with a toolrest holder. The toolrest holder shall have bearing surfaces on the bottom side of the base to provide horizontal adjustment along and lateral adjustment across the flat bedways. A single clamping or locking device shall retain the toolrest holder in adjusted position along the bedways. The top of the toolrest holder shall have a socket for accommodating the shank of standard toolrests. The socket shall hold the toolrest rigidly in place.

3.4.5.1 Hand feed carriage. When specified (see 6.2), type I and type II style 1 lathes shall be fitted with a hand feed carriage. The base of the carriage shall be supported by the flat ways and an inverted V-way on the

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front of the bed. A manually controlled pinion shall mesh with a rack gear mounted on the front of the bed. The top of the bed and the guide of the cross slide, supporting a component swivel tool post, shall have a scale marked in increments not greater than one sixteenth of an inch to facilitate cutting and turning workpieces to a definite length and depth.

3.4.6 Drive.

3.4.6.1 Type I, direct drive. The headstock spindle for type I lathes shall be direct motor (motor headstock) drive. The direct motor drive shall provide not less than four spindle speeds conforming to table I. A manually operated spindle speed selector shall be located on the operator's side of the headstock. Spindle speeds shall be permanently and legibly marked corresponding to each of the positions to which the spindle speed selector may be positioned.

3.4.6.2 Type II, belt drive. The spindle shall be belt driven from the motor through variable pitch pulleys, a variable pitch split sheave pulley or cone type pulleys to provide the step speed or infinite speed range in table I. The manual speed control, providing infinitely variable speeds, shall move a pointer across a graduated speed scale to indicate spindle speeds. The belt drive shall have means for belt alignment and maintaining belt tension.

3.4.7 Electrical system. The electrical system shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be wired to draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Unless otherwise specified (see 6.2), the lathe shall be initially wired for operation on 460 volts.

3.4.7.1 Motors. Motors shall meet the requirements of 3.4.7. Motors shall be rated for 60-minute duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.7.2 Hour meter. When specified (see 6.2), the lathe shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time for the main drive motor. The meter shall be of the non-resetting type and shall have a range of 0 to 9,999 hours in increments of not grater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the lathe.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the lathe shall meet the minimum requirements listed in table I for the machine specified.

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3.6 Alignment accuracy. Each lathe shall meet the alignment accuracies specified in table II.

3.7 Performance. Lathes equipped with a hand feed carriage shall be capable of meeting the following performance accuracies:

Facing (workpiece mounted on faceplate):
Faced surface shall be flat within 0.005 inch per foot of workpiece diameter.

Between centers turning:
Out-of-round - Not to exceed 0.003 inch.
Taper - Not to exceed 0.003 inch per foot.

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.10 Repair parts. Required repair parts shall be furnished as specified and fully described (see 6.2).

3.11 Marking on instruments, control panels, charts,and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Servicing interval
Type of lubricant
Federal or military specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall provide the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)

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Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Workmanship. Workmanship of the lathe and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

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4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the first article examination and all test to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4. 8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. The lathe shall be examined to determine compliance with all requirements of this specification.

4.7 Test.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 to 104 degrees Fahrenheit and 20 percent to 95 percent relative humidity.

4.7.2 Operational test. The lathe shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Alignment accuracy tests. The lathe shall be analytically tested for conformance with the alignment accuracy requirements of table II. The test methods and instruments used for the alignment accuracy tests shall be at the supplier's option provided that they accurately measure the lathe's alignment.

4.7.4 Performance test. The lathe shall perform the following test. The instruments and methods used in performing the test shall be at the option of the contractor provided the spindle speeds are accurately determined.

4.7.4.1 Headstock spindle speed test. Headstock spindle speed shall be tested by meter capable of registering RPM from a low of 250 or less, to a high of 3650 or more. Each spindle speed setting or the lowest and highest speed of the variable speed range shall be tested under no load. The spindle speed shall be within the speed requirements specified in table I.

4.7.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The lathes covered by this specification are intended for use in shops where woodworking operations are performed.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. Gap bed, if required (see 3.4.2).

k. Hand feed carriage, if required (see 3.4.5.1).

l. Electrical system, if different (see 3.4.7).

m. Initial voltage, if different (see 3.4.7).

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n. Motors, if different (see 3.4.7.1).

o. Hour meter, if required (see 3.4.7.2).
p. Size and capacity, if different (see 3.5).

q. Specify and fully describe optional equipment, if required (see 3.9).

r. Specify applicable repair parts, if required (see 3.10).

s. Lubrication plate, if required (see 3.11.1).

t. Information to be provided on lubrication plate, if different (see 3.11.1).

u. Nameplate, if different (see 3.11.2).

v. Inspection conditions, if different (see 4.3).

w. First article inspection, if different (see 4.4).

x. Quality conformance inspection, if different (see 4.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.13	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
3.13	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language

The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

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6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-376).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

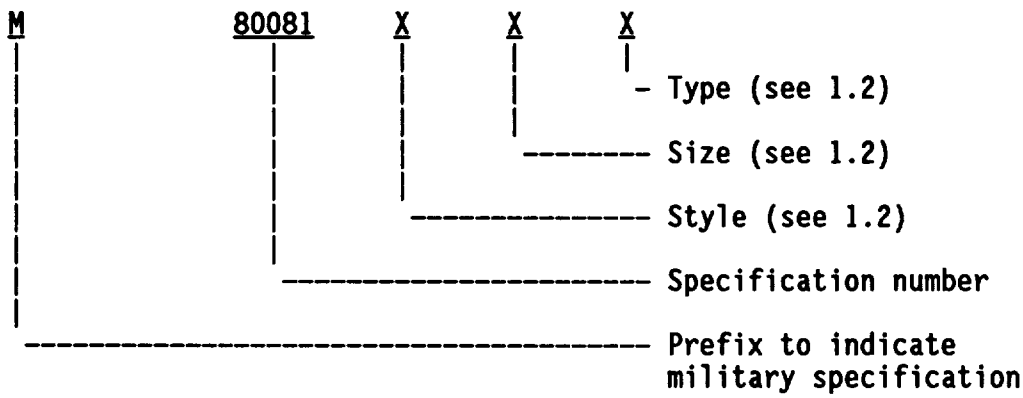
6.8 Training. Training required by the procuring activity should be provided as specified in the contract.

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6.9 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.10 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.11 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.12 Subject term (keyword) listing.

Boring
Facing
Turning

6.12 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I

Sizes and capacities

	Characteristics	Type I	Type II		
			Size 12		Size 14
			Style 1	Style 2	Style 1
	Swing diameter,(inches)				
	Over bedway	12	12	12	14
	Over carriage	9-1/2	9-1/2	-	11-1/2
*	Thru gap (diameter & width)	-	16x3	16x2-1/2	-
	Distance between centers (inches)	36	36	36	36
**	Cross slide traverse (inches)	5-1/2	5-1/2	-	5-1/2
**	Compound slide traverse (inches)	3-1/2	3-1/2	-	3-1/2
	Number of spindle speed selections	4	infinite	4	infinite
	Spindle speeds (RPM +/-200	600,1200 1800,3600	800 to 2100	915,1380 2150,3260	900 to 2700
	Motor Horsepower				
	Standard	3/4	3/4	1/2	1
	Optional	-	1	3/4	-

NOTE: Requirements are not less than those shown. When a range is shown the required performance is from the smaller figure or less to the larger figure or more.

* When the fixed gap bed requirement is specified in paragraph 3.4.2.
** When hand feed carriage is specified in paragraph 3.4.5.1.

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TABLE II
Alignment Accuracy

Test	Permissible error
Center runout (center installed in spindle taper; measured perpendicular to taper at center of taper)	0.002 inch, TIR
Spindle internal taper runout (measured on precision test bar inserted in spindle taper; measured 3" from spindle end)	0.006 inch, TIR
Faceplate runout, axial (measured on face of faceplate, at outer diameter)	0.005 inch, per foot
Spindle axis parallel to bedways (measured in both vertical and horizontal planes)	0.005 inch per foot

Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activities:
Army - AV, CE
Navy - YD, MC

Preparing activity:
DLA - IP

Project (3220-0164)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-80081D	900814

3. DOCUMENT TITLE

Lathes, Woodworking

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSM	(1) Commercial (901) 775-6257	(2) AUTOVON AV 683-6257
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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